

# Design of a GaN Half-Bridge Building Block for a Stacked Poly-Phase Bridge: EPC2361, 75 V / 6.25 kW

SPB Building-Block Design Toolkit  
September 11, 2026

## Abstract

This paper reports the electrical and thermal design of one half-bridge building block of a stacked poly-phase bridge, using the EPC2361 100 V eGaN HEMT. Gate turn-on and turn-off resistance are optimized independently against a full 120-point LTSpice double-pulse characterization spanning device current, gate resistance, and power-loop inductance, run against the vendor’s own SPICE model. The recommended point — **N=4 devices in parallel per switch position,  $R_{g,on}=0.5\ \Omega$ ,  $R_{g,off}=2.7\ \Omega$ ,  $f_{sw} \approx 99.3\ \text{kHz}$**  — meets a 99.5% full-load efficiency target and a 125 °C junction-temperature ceiling with margin, using thermal and power-loop parameters grounded in three independent published GaN half-bridge designs rather than assumed values.

## 1 Design Specification

One building block = one half-bridge leg of the stacked poly-phase bridge, two switch positions each populated by  $N$  EPC2361 devices in parallel. Table 1 summarizes the target; Table 2 the device parameters used, sourced from the EPC2361 datasheet (rev. 2.4) and its companion FEA thermal model unless noted.

Table 1: Design targets.

|                               |                        |                       |
|-------------------------------|------------------------|-----------------------|
| $P_{out}$ (this leg)          | 6250 W                 | <sup>a</sup> device’s |
| $V_{dc} / V_{rms, fund}$      | 75 V / 30 V            |                       |
| $\cos \varphi / M$ (SVM)      | 0.95 / 1.15            |                       |
| Efficiency target (full load) | 99.5%                  |                       |
| $T_J$ ceiling                 | 125 °C <sup>a</sup>    |                       |
| Max. $V_{DS}$ overshoot       | 20% of $V_{dc}$ (15 V) |                       |
| $T_{ambient}$                 | 45 °C                  |                       |
| Dead time                     | 10 ns                  |                       |

recommended max for full utilization, used directly (not the 150 °C absolute max less a margin).

Table 2: EPC2361 parameters (see §2.3 for the  $R_{\theta JC}$  note).

|                                 |                                                        |
|---------------------------------|--------------------------------------------------------|
| $R_{DS(on), 25} / k_T$          | 0.75 m $\Omega$ / $6.4 \times 10^{-3}\ \text{°C}^{-1}$ |
| $Q_{GS} / Q_{GD} / Q_G$         | 8.5 / 3.8 / 34 nC                                      |
| $R_{g, int}$                    | 0.4 $\Omega$                                           |
| $V_{SD0} / R_{SD}$              | 1.6 V / 5 m $\Omega$                                   |
| $R_{\theta JC} / R_{\theta JB}$ | 0.24 / 1.5 $\Omega \cdot \text{W}^{-1}$                |

## 2 Methodology

### 2.1 Conduction and switching loss

Conduction loss uses the closed-form result that, for a gated-on GaN channel conducting symmetrically in both current directions (verified against the EPC2361

SPICE model’s symmetric `bswitch` behavioral source), synchronous-rectification conduction loss per switch position is *independent of modulation index and power factor*:  $P_T = R_{DS(on)}(T_J) I_{pk}^2 / 4$ , plus a dead-time third-quadrant term using  $V_{SD}(I) = V_{SD0} + R_{SD}|I|$ . Full derivation and proof: `docs/derivations.md`.

Switching loss is obtained by numerically averaging  $E_{on}(I_D) + E_{off}(I_D)$  over the fundamental-cycle current waveform and multiplying by  $f_{sw}$ .  $E_{on}$  and  $E_{off}$  come from a clamped-inductive double-pulse LTSpice test against the real EPC2361 subcircuit (`EPCGaNLibrary.lib`), with the complementary device providing the freewheel path via its own third-quadrant conduction (no ideal diode substituted).

### 2.2 Independent $R_{g,on}/R_{g,off}$ optimization

The double-pulse gate-drive network routes turn-on and turn-off current through electrically separate diode/resistor paths, so  $E_{on}(I_D, R_{g,on}, L_{loop})$  and  $E_{off}/\text{overshoot}(I_D, R_{g,off}, L_{loop})$  were characterized in two *decoupled* sweeps (120 LTSpice runs total: 4 currents  $\times$  5 gate resistances  $\times$  3 loop inductances, per sweep), each holding the other gate resistor at a 2  $\Omega$  reference.  $V_{DS}$  overshoot is governed by the turn-off edge only, so:

- $R_{g,off}$  is bisected for the smallest value meeting the 15 V overshoot limit (overshoot decreases monotonically with  $R_{g,off}$ , confirmed in the sweep data);
- $R_{g,on}$  is set to the search floor, since  $E_{on}$  increases monotonically with  $R_{g,on}$  in every sweep point and carries no overshoot penalty in this model.

Real designs sometimes deliberately raise  $R_{g,on}$  above  $R_{g,off}$  regardless (e.g. Cooke & Rogers [3] use 15  $\Omega$ /2  $\Omega$  source/sink) to limit turn-on  $dv/dt$  and the risk of Miller-induced parasitic turn-on of the complementary device – a failure mode not modeled here; a crosstalk check against the real gate driver and PCB is still required before finalizing  $R_{g,on}$  on hardware.

### 2.3 Power-loop inductance and thermal parameters

Loop inductance uses  $L = \mu_0 h l / w$  (overlapping-plane/PCB-trace formula, derived from Ampère’s law). This design’s geometry ( $h=0.2\ \text{mm}$ ,  $l=15\ \text{mm}$ ,  $w=8\ \text{mm}$ ) gives 0.47 nH – validated against Cooke & Rogers [3], who measure/estimate 0.34–0.74 nH with the *same formula* on a comparable 4-layer PCB half-bridge. The double-pulse sweep spans  $L_{loop} \in \{0.3, 0.7, 1.5\}\ \text{nH}$  to

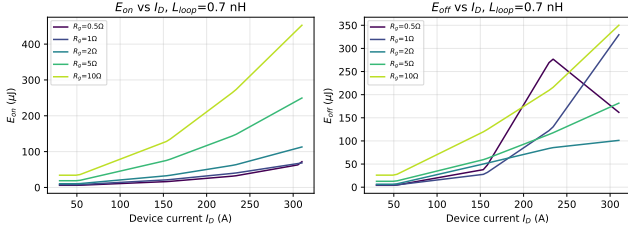


Figure 1:  $E_{on}$ ,  $E_{off}$  vs. device current for  $R_g \in \{0.5, 1, 2, 5, 10\} \Omega$  at  $L_{loop}=0.7$  nH (LTSpice double-pulse data, EPC2361).

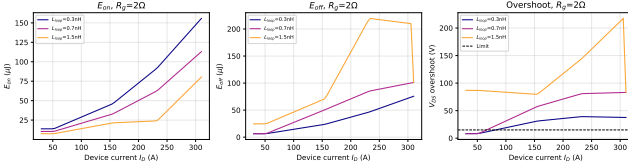


Figure 2:  $E_{on}$ ,  $E_{off}$ , and  $V_{DS}$  overshoot vs. device current for  $L_{loop} \in \{0.3, 0.7, 1.5\}$  nH at the  $2 \Omega$  reference gate resistance.

bound this range and show sensitivity, rather than relying on one point estimate.

Thermal parameters were revised from a prior placeholder-heavy pass using three literature designs of the same device class:  $R_{\theta,int}$  (TIM) =  $2.5 \Omega \cdot W^{-1}$  per device, from Cooke & Rogers' [3] A1780 isolating TIM (0.8 mm compressed,  $k=17.8$  W/mK) scaled to EPC2361's  $\sim 15$  mm<sup>2</sup> footprint ( $R = t/kA \approx 3.0 \Omega \cdot W^{-1}$ , rounded down slightly for better compression);  $R_{\theta,sa}$  (heatsink) =  $0.6 \Omega \cdot W^{-1}$  for the assembly, informed by their  $0.9 \Omega \cdot W^{-1}$  60×60 mm forced-air finned heatsink (sized up for this design's larger total loss budget). Also cross-checked: the datasheet headline  $R_{\theta JC}=0.2 \Omega \cdot W^{-1}$  versus the companion FEA Foster-ladder sum ( $0.0247+0.0421+0.173=0.240 \Omega \cdot W^{-1}$ ) – a real  $\approx 20\%$  discrepancy, not rounding; the more detailed/conservative 0.24 value is used throughout.  $R_{\theta JB}$ 's ladder sum (1.51) matches its 1.5 headline to  $<1\%$ .

### 3 Results

Figures 1–2 show the full characterization: both  $E_{on}$  and  $E_{off}$  increase monotonically with device current and gate resistance at every loop inductance tested (the expected switching-speed/loss trade-off), while  $V_{DS}$  overshoot decreases with  $R_g$  and increases sharply with  $L_{loop}$  – at the largest tested loop inductance (1.5 nH) and lowest gate resistances, overshoot exceeds the device's rated  $V_{DS,tr}$  margin at high current, directly motivating a low-inductance layout rather than compensating for a poor one with gate resistance alone.

Figure 3 sweeps parallel-device count  $N$ ; for each  $N$ ,  $R_{g,off}$  is re-optimized against the overshoot limit and  $f_{sw}$  against the loss/thermal budget. Table 3 gives the numeric trade-off. Figure 4 shows loss composition vs.  $f_{sw}$  at the recommended point, confirming the search-bound ceiling discussed below.

Loss composition and max  $f_{sw}$  vs. parallel-device count

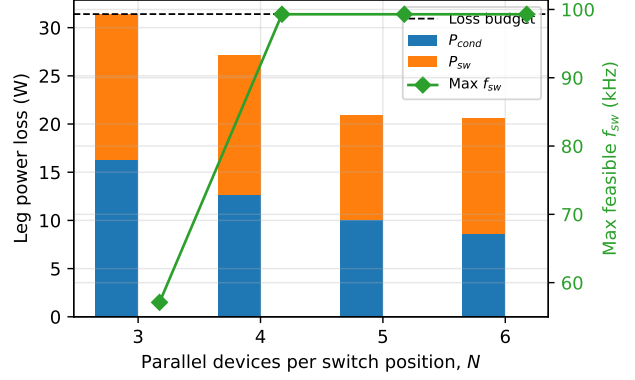


Figure 3: Loss composition and maximum feasible  $f_{sw}$  vs. parallel-device count  $N$ .

Loss vs.  $f_{sw}$  at  $N=4$ ,  $R_g=(0.5/2.7) \Omega$

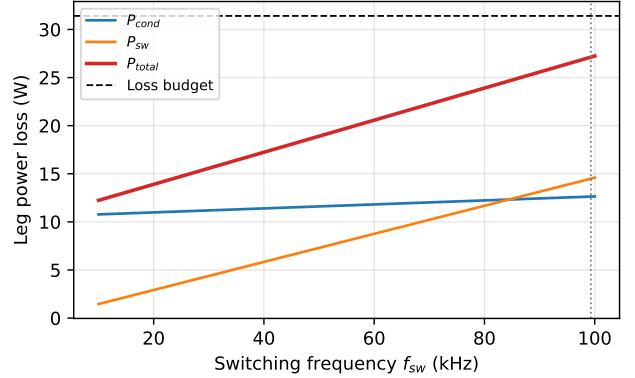


Figure 4: Loss vs. switching frequency at the recommended design point.

#### 3.1 Recommended design point

$N=4$ ,  $R_{g,on}=0.5 \Omega$ ,  $R_{g,off}=2.7 \Omega$ ,  $f_{sw} \approx 99.3$  kHz: loss 27.12 W of 31.41 W budget (86% used),  $T_J=62.4^\circ\text{C}$  ( $62.6^\circ\text{C}$  margin to the  $125^\circ\text{C}$  ceiling),  $V_{DS}$  overshoot 14.8 V.

$N=1$  and  $N=2$  are infeasible: even at the maximum searched  $R_{g,off}$ , overshoot exceeds the 15 V limit at their per-device currents – a conclusion from real double-pulse data, not a rule of thumb. Where the recommended  $f_{sw}$  sits at (or near) the search's configured ceiling with unused loss/thermal margin, that is a search-bound limit, not a physical one – raising  $f_{sw\_max\_Hz}$  and re-running would find the true maximum.

### 4 Discussion and Limitations

- **Even current sharing across  $N$  parallel devices is assumed**, not modeled. Palma et al. [5] show real boards see dynamic imbalance from device-to-device  $V_{th}$  spread; layout symmetry (e.g. the “butterfly” gate-loop structure of Wattenberg et al. [6], achieving  $<17\%$  gate-loop impedance spread across 8 parallel devices) is essential and not something this model can check.

Table 3:  $f_{sw}$  vs. parallel-device-count trade-off at  $L_{loop}=0.7$  nH (feasible points only).

| $N$      | $R_{g,on}$<br>( $\Omega$ ) | $R_{g,off}$<br>( $\Omega$ ) | $f_{sw}$<br>(kHz) | $P_{total}$<br>(W) | $T_J$<br>( $^{\circ}\text{C}$ ) | Margin<br>( $^{\circ}\text{C}$ ) | Ovs.<br>(V) |
|----------|----------------------------|-----------------------------|-------------------|--------------------|---------------------------------|----------------------------------|-------------|
| 3        | 0.5                        | 6.9                         | 57.1              | 31.32              | 68.7                            | 56.3                             | 14.8        |
| <b>4</b> | <b>0.5</b>                 | <b>2.7</b>                  | <b>99.3</b>       | <b>27.12</b>       | <b>62.4</b>                     | <b>62.6</b>                      | <b>14.8</b> |
| 5        | 0.5                        | 0.6                         | 99.3              | 20.97              | 57.0                            | 68.0                             | 14.7        |
| 6        | 0.5                        | 0.6                         | 99.3              | 20.60              | 55.9                            | 69.1                             | 8.5         |

- **Steady-state, full-load thermal only** – no transient/duty-cycled  $T_J$  model (the FEA Foster ladder is recorded in `data/devices/EPC2361.yaml` for future extension).
- **Loop inductance** is bounded, not pinned to a single assumed value, by sweeping  $L_{loop}$  – but all three values are analytical/literature estimates until measured on real hardware.
- **Crosstalk/false-turn-on** is not modeled;  $R_{g,on}$  is chosen purely to minimize switching loss, which may be more aggressive than a real gate-driver’s pull-down strength and PCB layout can safely support (§2.3).

## References

- [1] EPC2361 datasheet, rev.2.4, Efficient Power Conversion Corp., 2026.
- [2] EPC2361 Thermal SPICE Model, Efficient Power Conversion Corp.
- [3] M. W. J. Cooke, D. J. Rogers, “120 V, 155 A Traction Inverter Using Eight Parallel GaN HEMTs,” 2025/2026.
- [4] J. A. Brothers, T. Beechner, “GaN Module Design Recommendations Based on the Analysis of a Commercial 3-Phase GaN Module,” IEEE ECCE, 2019.
- [5] M. Palma, V. Barba, S. Musumeci, “Parallel Connection of GaN FETs: an Experimental Investigation Approach,” PCIM Europe, 2024.
- [6] M. Wattenberg, O. G. Lorenz, J. Sanchez, “A Multi-Kilowatt Low-Profile GaN Inverter for Light Electric Vehicles and High-Power Tools,” IEEE APEC, 2023.
- [7] F. Casanellas, “Losses in PWM inverters using IGBTs,” IEE Proc. Electr. Power Appl., 1994.